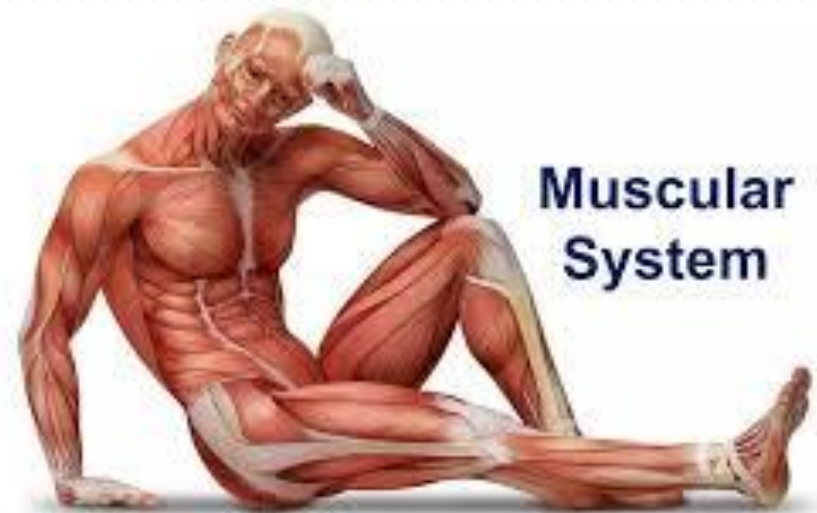


Management of patient with Musculoskeletal diseases



**Muscular
System**

Dr . Bardies Shaban



الحديث الجانبي يشتت الانتباه



من فضلك اغلق الهاتف

4- Osteomyelitis & Osteoporosis

- ◉ Definition
- ◉ Causes
- ◉ Manifestations
- ◉ Diagnosis
- ◉ Complications
- ◉ Managements

Infection and **inflammation** of the **bone** or **bone marrow**, usually caused by bacteria ,fungi or other germs that spread from the bloodstream or nearby tissue.

Osteomyelitis



Causes Bacterial infection
Staphylococcus aureus

Open fractures or surgery

Infected prosthetic joints

Spread from nearby infections
(e.g., cellulitis)

Bloodborne spread
(hematogenous)

Manifestations

OSTEOMYELITIS



Pain



Fever and chills



Redness and swelling in affected area



Lack of energy



Irritability



Drainage or pus



Inability to move the affected limb

complications

Bone necrosis (death of bone tissue)

Sepsis

Impaired growth (in children)

Amputation

Management

- **Surgical :**

To remove necrotic bone or drain abscess

- **Medical :**

Antibiotics

Pain management

Immobilization

Hyperbaric oxygen therapy

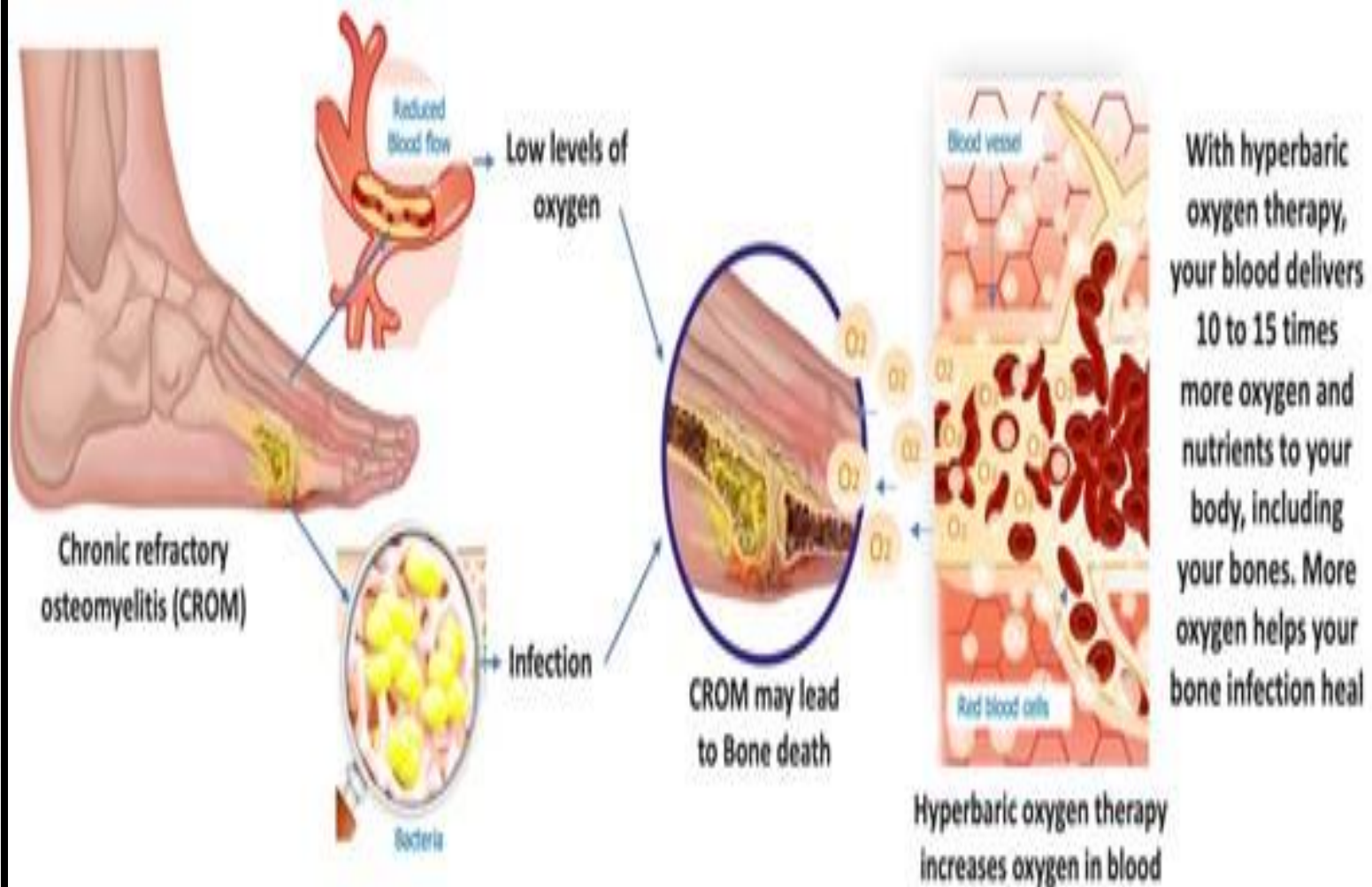
Hyperbaric oxygen therapy : closed space with oxygen higher than in normal air pressure. It provide the body tissues with pure oxygen



It may be individualized for **one patient** , or **group** therapy



Hyperbaric oxygen therapy for Osteomyelitis



Nursing process

1- Assessment :

- Monitor for fever, pain, swelling, and wound drainage
- Check lab results (WBC, CRP)
- Assess for risk factors (diabetes, recent surgeries)

2- Nursing diagnosis :

- Acute pain related to infection
- Impaired physical mobility related to pain
- Risk for sepsis
- Risk for impaired skin integrity

Nursing process

3- planning :

- Manage pain
- Reduce infection
- Prevent complications
- Promote mobility

4- implementation :

- Administer antibiotics as prescribed
- Wound care
- Educate on medication adherence and signs of infection
- Promote rest and immobilization

Nursing process

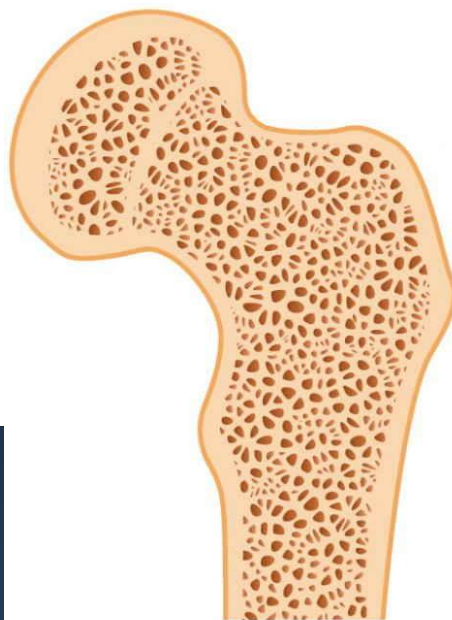
5- Evaluation :

- Reduction in signs of infection
- Decreased pain
- Improved mobility
- No complications (as sepsis)

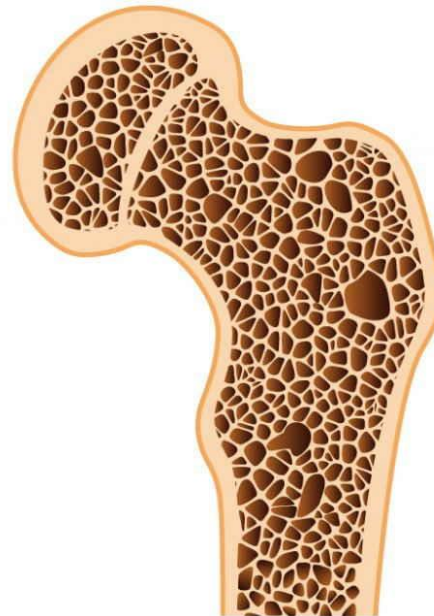
Osteoporosis

Definition

Decrease **bone mass** and **deterioration** of bone tissue, leading to **fragile bones** and increased risk of **fractures**.



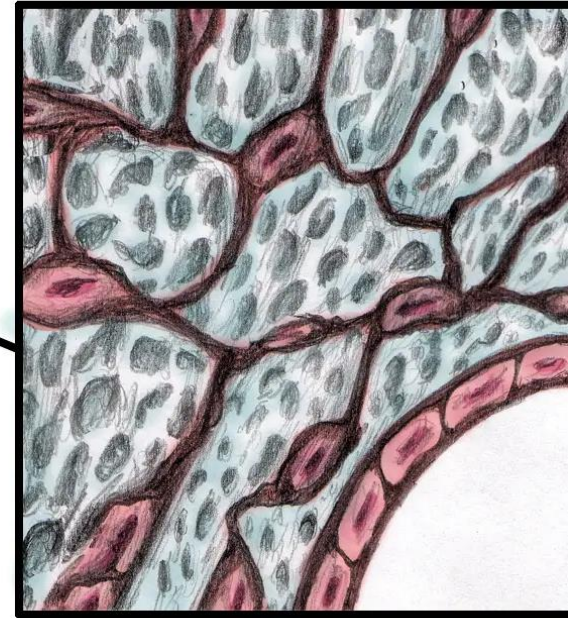
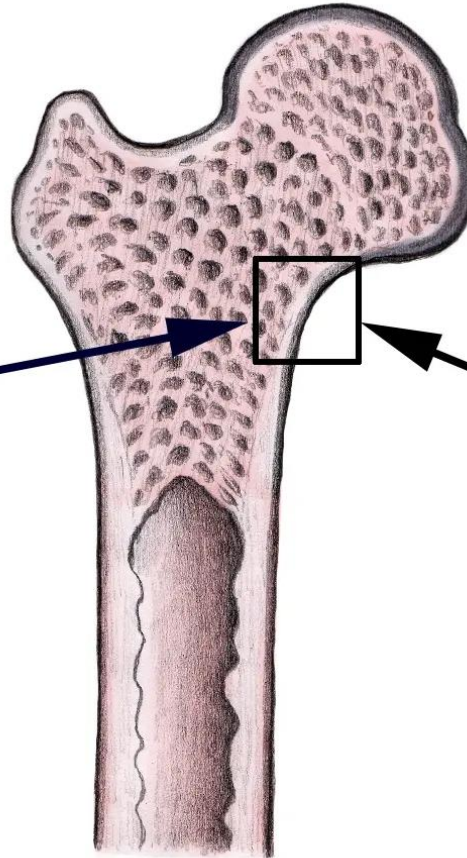
Healthy bone



Osteoporosis



Section through bone
with osteoporosis



Section through normal bone

Causes

Aging
(postmenopausal women)

Long-term use of corticosteroids

Hormonal changes
(↓ estrogen or testosterone)

Malnutrition
(low calcium, vitamin D)

Sedentary lifestyle

SYMPTOMS OF OSTEOPOROSIS



**CONSTANT
BACK PAIN**



**LOSS OF
HEIGHT**



**ABNORMALLY
STOOPED POSTURE**



**BONES THAT BREAK
EASILY ON THE
SLIGHTEST TRAUMA**



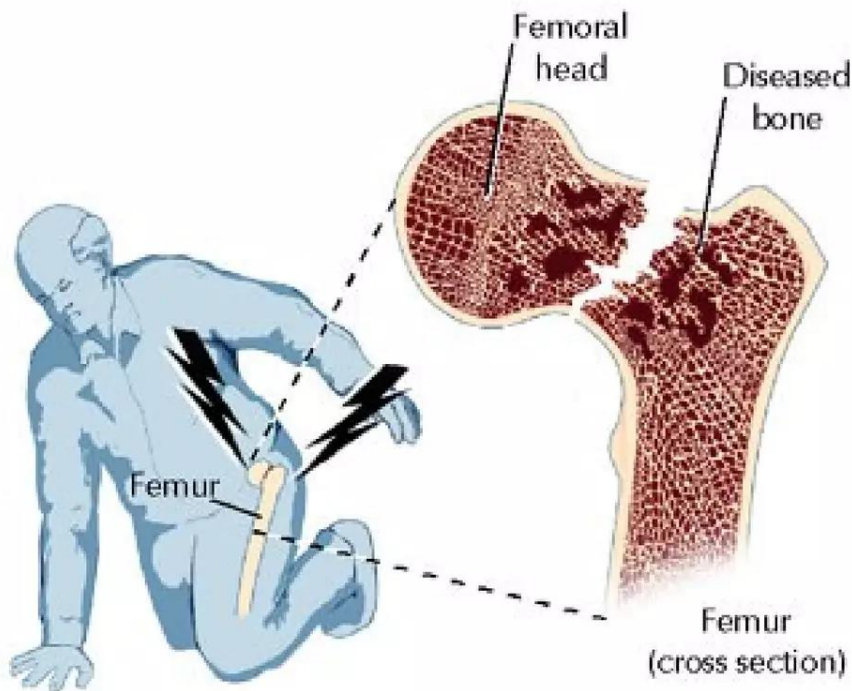


Complications

FRACTURE ,

The most serious complication of Osteoporosis that leads to

- **Increased morbidity**
- **Increased mortality**
- **Decreased quality of life**



What does an osteoporosis treatment plan include?



Medication



Healthy diet



Calcium and
vitamin D



Exercise



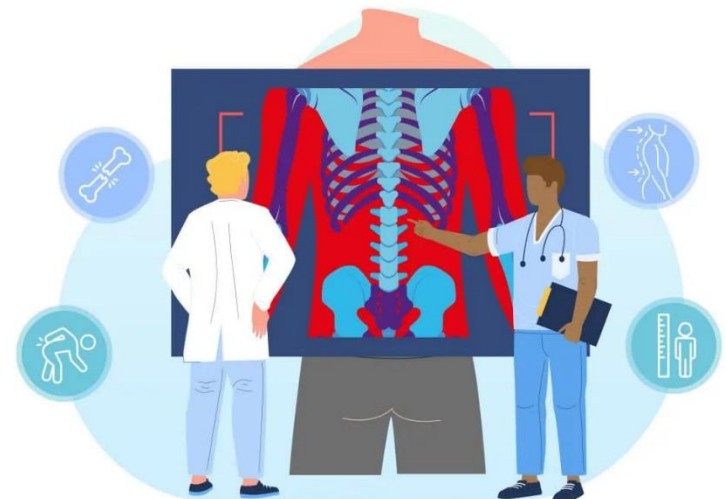
Prevent falls



Nursing process

1- Assessment

- Assess for **risk factors** (age, diet, lifestyle)
- Review bone mineral density (**BMD**) results
- Monitor for signs of **fractures**



Nursing process

2- Diagnosis

Risk for injury related to fragile bones

Impaired physical mobility

Knowledge deficit about disease management



Nursing process

3- Planning

Prevent

Fractures

Improve

Bone health

Educate on

lifestyle changes



Nursing process

4- Implementation

- Fall prevention
- Administer medications as prescribed
- Encourage calcium-rich diet and exercise
- Educate on medication adherence



Nursing process

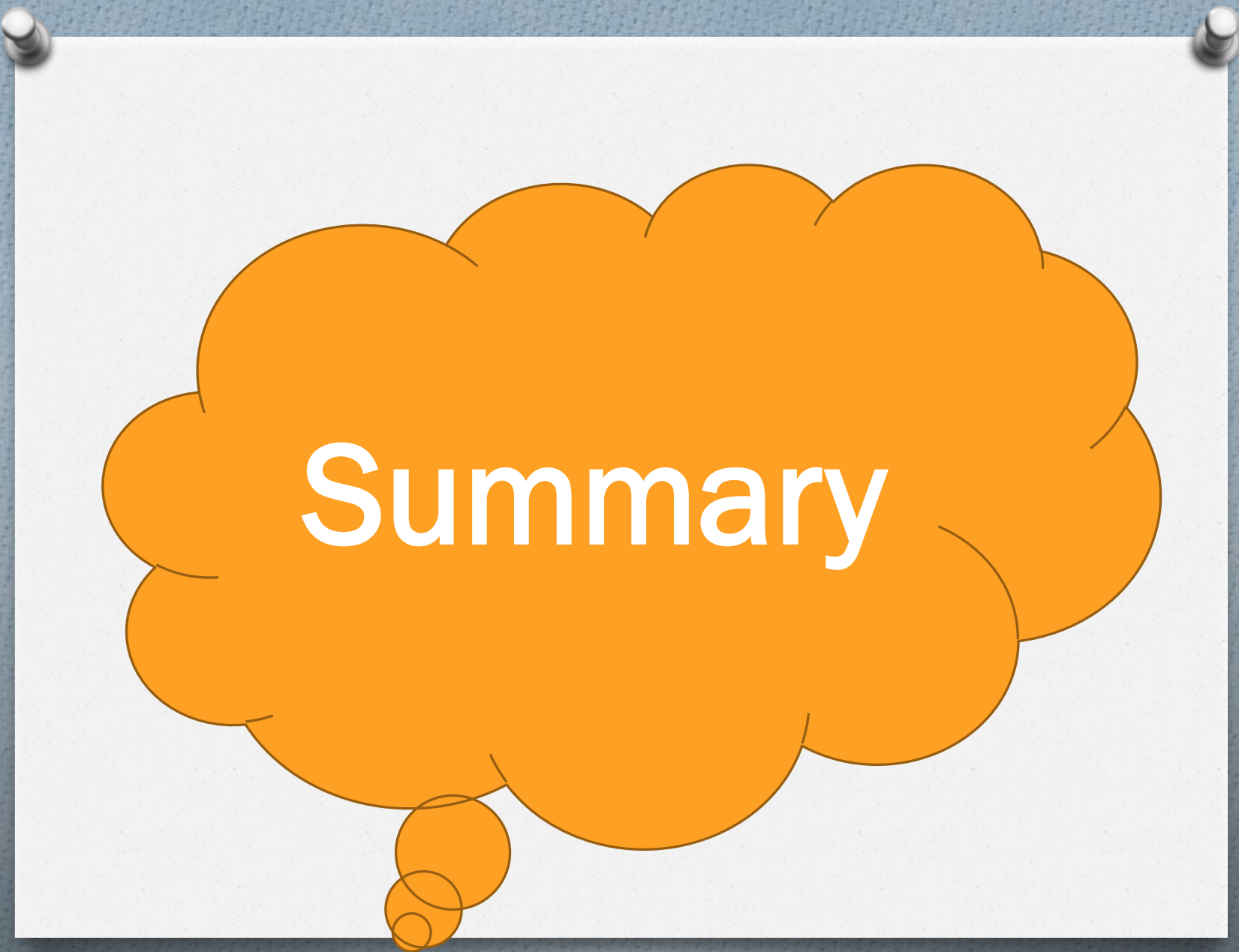
5- Evaluation

No new fractures

Improved BMD

Increased knowledge of disease and self-care





Summary



- **List 3 causes of osteomyelitis**
- **List 3 causes of osteoporosis**
- **List 4 manifestations of osteoporosis.**



Thank
you